First Demonstration of BEOL-Compatible ALD InGaO High-Voltage TFTs

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***Abstract*— In this work, we report the first experimental demonstration of atomic-layer-deposited (ALD) InGaO (IGO) high-voltage TFTs using a drain offset structure with back-end-of-line (BEOL) compatibility (< 300 °C). Using ALD Al-doped HfO_2 as a bottom-gate dielectric and IGO as a channel, a high carrier mobility (μ_{FE}) of 35 $cm^2/V{\cdot}s$ and a high on/off ratio of 10^9 with a reasonable threshold voltage (V_{th}) of −2 V were demonstrated by optimizing the O_2 annealing time. The high-voltage IGO TFT with a gate-to-drain offset length of 20 μm possesses a record high breakdown voltage (V_{BD}) of 638 V and Baliga's Figure of Merit (BFOM) of 2.4 MW/cm^2. The trade-off relationship between V_{BD} and the specific on-resistance of ALD high-voltage IGO TFTs surpasses the theoretical limit of the single-RESURF lateral double-diffused MOSFET (LDMOS).**

***Index Terms*— Atomic layer deposition, indium gallium oxide (IGO), high-voltage, thin film transistors.**

I. Introduction

HIGH-VOLTAGE thin film transistors (HVTFTs) are promising devices for driving field-emitter arrays, large area direct detection X-ray imaging and microelectromechanical system applications [1], [2], [3]. Previous research has reported HVTFTs based on amorphous silicon, amorphous oxide semiconductors (AOSs) and MgZnO [4], [5], [6]. Among these channel materials, AOSs are highly competitive due to their large bandgap, high mobility, and low thermal budget for growth and fabrication [7], [8]. A wide bandgap is beneficial for achieving a high breakdown electric field and low leakage current in HVTFTs. Moreover, the moderate mobility and high E_C in AOSs potentially offer a better trade-off between specific on-resistance ($R_{on,sp}$) and V_{BD} [9]. Recently, high-voltage transistors have been studied using an amorphous TFT with a V_{BD} exceeding 400 V via magnetron sputtering [10]. Sputtering is a conventional technique with good uniformity, and productivity and low-temperature deposition capabilities and has been widely used in depositing AOSs [11]. However, it suffers from device-to-device variation and reliability issues. In addition, the high growth rate and the targets with fixed film composition in the sputtering system limit its application in advanced channel designs [12]. On the other hand, ALD can achieve accurate thickness control, high conformality, and wafer-scale uniformity, which makes ALD a viable alternative to the sputtering process for depositing AOSs [13]. High-performance indium oxide or doped-indium oxide transistors based on ALD channels have been demonstrated in recent years [14], [15], [16], [17]. Among this family of materials, IGO is a promising material with a higher oxygen dissociation energy and thermal stability than pure In_2O_3 because of its lower carrier concentration. In addition, the introduction of Ga would induce minimal distortions to the In_2O_3 host structure and possibly maintain a higher mobility than other metal cation dopants [18]. However, high-performance ALD high-voltage IGO TFTs have not yet been reported.

In this work, we experimentally report the first high-performance IGO HVTFTs obtained via ALD. The IGO bottom-gate HVTFTs exhibit a high carrier mobility of 35 $cm^2/V{\cdot}s$, an on-state current of 794 μA, and a low subthreshold swing (SS) of 90 mV/dec. A record high V_{BD} of 638 V and a BFOM of 2.4 MW/cm^2 were also demonstrated. This study demonstrated that ALD oxide HVTFTs are promising candidates for high-voltage and high ON-current TFTs.

II. Device Fabrication

A schematic diagram of a bottom-gate InGaO transistor is shown in Fig. 1(a) and the corresponding fabrication process flow is illustrated in Fig. 1(b). First, Ni/Au (20/40 nm) was deposited by e-beam evaporation using a lift-off process, followed by ALD of 15 nm HfAlO (10% Al) films as gate dielectrics at 250 °C. A 14-nm IGO film was deposited by ALD at 250 °C. Notably, TMIn and H_2O are used as In and O precursors for In_2O_3 subcycles and TMGa and ozone are used as Ga and O precursors for Ga_2O_3 subcycles. X-ray photoelectron spectroscopy (XPS) revealed that the atomic ratio of In to Ga was 67:33. The channel region was defined using photolithography followed by wet etching with 1:1 hydrochloric acid. Source and drain contacts with Ni/Au (30/70 nm) were formed by e-beam evaporation. Ni metal materials are used as contact materials due to their low contact resistance [7]. Finally, all the devices were annealed in O_2 at

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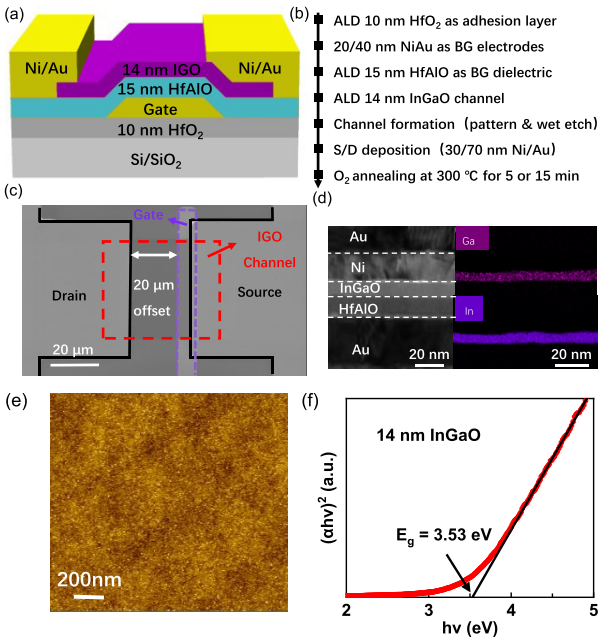


Fig. 1. (a) Schematic view of a bottom-gate HV IGO TFT. (b) Fabrication process of back-gate high-voltage IGO transistors. (c). SEM image of a bottom-gate HV IGO TFT with a L_{offset} of 20 μm. (d) Left part: the HRTEM image clearly shows a sharp interface with thin dielectrics and channels. Right part: EDS mapping of Ga and In. (e) AFM measurement of the surface roughness of a 14 nm as-grown IGO film. (f) $(\alpha h\nu)^2$-hv plot of IGO films. The bandgap of IGO is 3.53 eV.

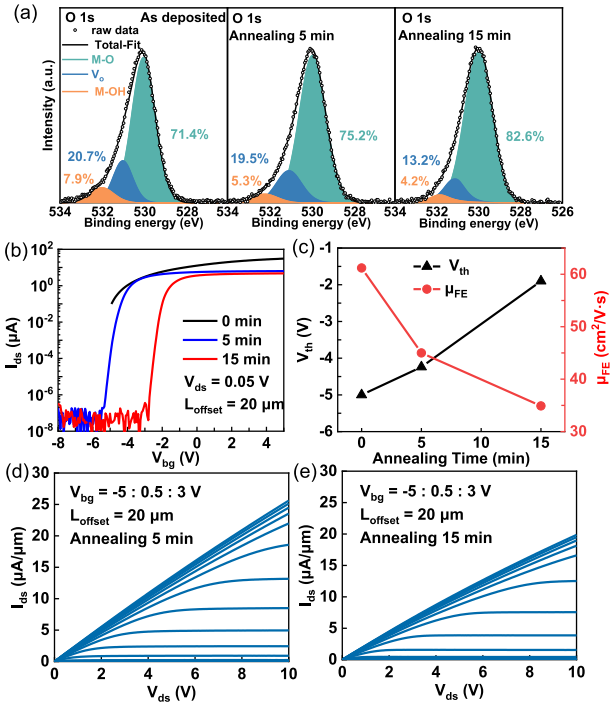


Fig. 2. (a) XPS spectra of deconvoluted O 1 s peaks for IGO films without and with O_2 annealing. (b) I_{ds}-V_{bg} curves of IGO with L_{offset} = 20 μm and V_{ds} = 0.05 V with O_2 annealing at different times. (c) V_{th} and μ_{FE} as a function of annealing time. As expected, increasing the annealing time can shift V_{th} positively and decrease the μ_{FE}. Output characteristics of HV IGO TFTs after O_2 annealing for (d) 5 min, and (e) 15 min.

300 °C and the effect of the annealing time on the electrical performance was investigated. Electrical measurements of the InGaO HVTFTs were carried out at room temperature using a Cascade Tesla power probe station and an Agilent semiconductor parameter analyzer B1505A. Fig.1(c) shows a top-view scanning electron microscopy (SEM) image of a typical IGO HVTFT with a gate-to-drain offset length (L_{offset}) of 20 μm. The channel width is 40 μm and the gate length is 5 μm. Fig.1(d) shows a high-resolution cross-sectional TEM image of the IGO TFTs, where the thicknesses of HfAlO and IGO were confirmed to be 15 nm and 14 nm, respectively. Fig. 1(e) shows a low surface roughness of 0.36 nm for the IGO film measured by atomic force microscopy (AFM). The optical bandgaps (E_g) of the IGO films were calculated from the optical transmittance measurements by extrapolating the straight-line segments of the plot of $(\alpha h\upsilon)^2$ versus $h\upsilon$ to α = 0, as shown in Fig. 1(f), where α, h, and υ are the absorbance, Planck constant, and light frequency, respectively. The E_g of the IGO film is 3.53 eV.

III. Results and Discussion

To evaluate the effect of O_2 annealing on the IGO film, XPS was performed as shown in Fig. 2(a). Ar^+ sputter etching is used prior to XPS analyses with the intention of removing surface oxides and contaminants. The O 1 s spectrum of the IGO film can be divided into three subpeaks, which are attributed mainly to oxygen in the lattice (M-O), oxygen vacancies (V_O), and impurities such as hydroxide (O-H) [19]. This finding shows that increasing the O_2 annealing time could effectively reduce the concentrations of V_O and O-H [20]. Fig. 2(b) shows the transfer characteristics of typical high-voltage IGO TFTs with L_{offset} = 20 μm at V_{ds} = 0.05 V with different O_2 annealing times. The IGO device without annealing exhibited a low on/off ratio due to the presence of abundant oxygen vacancies and H doping in the channel during the ALD process. The μ_{FE} is extracted by $\mu_{FE} = L_{ch}g_m/(W_{ch}C_{ox}V_{ds})$, where g_m is the transconductance at small V_{ds}, W_{ch} is the channel width, and C_{ox} is the dielectric capacitance. With increasing annealing time from 5 min to 15 min, V_{th} increases and the μ_{FE} decreases from 45 $cm^2/V{\cdot}s$ to 35 $cm^2/V{\cdot}s$ as shown in Fig. 2(c). An ultralow SS of 90 mV/dec is achieved for the IGO device with O_2 annealing for 15 min. The low SS indicates a low interface trap density D_{it} value of $\sim 1.6 \times 10^{12}$ $cm^{-2}eV^{-1}$ using $SS = 60$ mV/dec $(1 + qD_{it}/C_{ox})$. The corresponding output characteristics of the devices with annealing are shown in Figs. 2(d) and 2(e). For the 15 min annealed IGO device, the maximum on-state current I_{on} is as high as 794 μA (19.8 μA/μm) at a gate voltage of 3 V and a V_{ds} of 10 V. In addition, the IGO film after 15 min of annealing has the smallest V_{th} (-2 V). This indicates a low electron concentration in the channel without bias voltage, which can effectively alleviate electric field crowding and improve the breakdown characteristics [10].

Figs. 3(a) and (b) show the breakdown characteristics of the ALD IGO TFTs after 15 min of annealing at L_{offset} = 2 μm and 20 μm, respectively. The saturation currents decrease due to the increase in drain offset resistance. The rapid increase in I_{ds} when V_{ds} was greater than $\sim$ 400 V in Fig. 3(b) can be attributed to a negative shift in V_{th} during the high-voltage measurement caused by self-heating and the high electric field [21], [22]. Fig. 3(c) shows the I_{ds} - V_{ds} performance degradation under V_{ds} sweep as a function of L_{offset}. The V_{BD} of the 2, 5, 10, and 20 μm drain offset IGO TFTs

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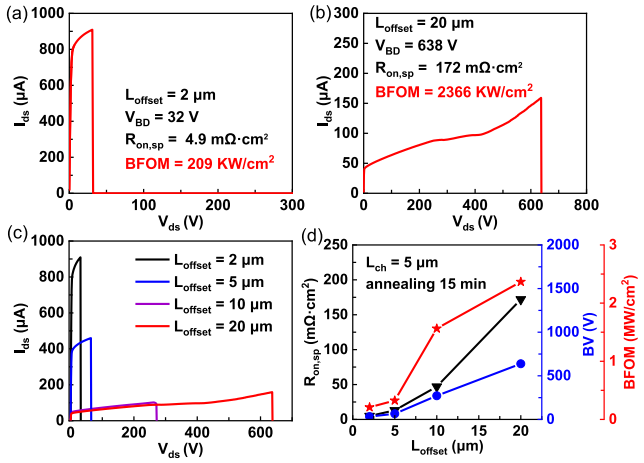


Fig. 3. Breakdown voltage measurement results of IGO HVTFTs with (a) $L_{offset} = 2\ \mu m$, and (b) $L_{offset} = 20\ \mu m$. (c) On-state current I_{ds} as a function of breakdown voltage for IGO HVTFTs with different L_{offset} values. (d) $R_{on,sp}$, V_{BD}, and BFOM as a function of L_{offset}.

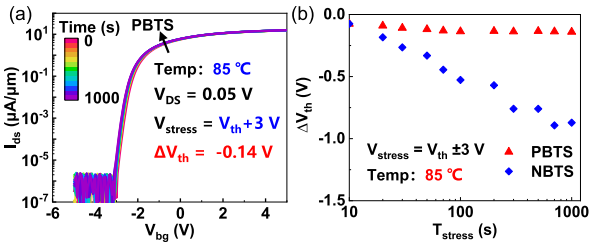


Fig. 4. (a) Transfer curve evolution of HV IGO TFTs during the PBTS test. (b) Time evolution of ΔV_{th} of HV IGO TFTs under PBTS and NBTS tests up to 1000 s.

are 32, 65, 271, and 638 V, respectively. The breakdown usually occurs near the drain with a high degree of electric field crowding. Increasing the distance from the drain to the gate can effectively reduce the electric field congestion near the drain and increase the breakdown voltage [23]. Fig. 3(d) presents the effect of L_{offset} on the $R_{on,sp}$, V_{BD}, and BFOM of IGO HVTFTs. As expected, $R_{on,sp}$ and V_{BD} increase as L_{offset} increases. Baliga's Figure of Merit (BFOM $= V_{BD}^2/R_{on,sp}$) was obtained to evaluate the performance of the HV IGO TFTs. As L_{offset} increases, the BFOM increases and reaches its maximum at an L_{offset} of 20 μm. The ALD IGO HVTFT with an L_{offset} of 20 μm achieves a record high V_{BD} of 638 V, a low $R_{on,sp}$ of 172 mΩ·cm^2, and a high BFOM of 2.4 MW/cm^2 [10], [24].

Fig. 4 shows the bias stress stability of IGO HVTFTs under a positive/negative bias temperature stability (PBTS/NBTS) of V_{th} −3/+3 V for 1000 s at 85 °C. A relatively small ΔV_{th} of −140/−870 mV under PBTS/NBTS for 1000 s is achieved. The donor-like traps, such as oxygen vacancies and hydrogen states could be ionized under high electric field, releasing electrons into the conduction band and leading to a negative V_{th} shift [25]. During the PBTS test, electrons accumulate at the channel/dielectric interface and a number of generated donor traps can be filled by these accumulated electrons, leading to a small negative V_{th} shift. However, electrons were depleted during the NBTS test and these generated donor traps remain positively charged, resulting in a more pronounced V_{th} shift [18]. The stability of ALD IGO TFTs

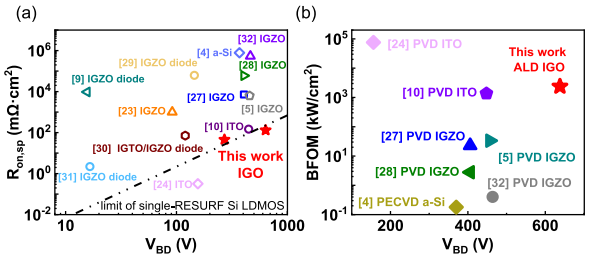


Fig. 5. (a) Benchmark of $R_{on,sp}$ versus V_{BD} of the HV IGO TFTs in this work, previously reported AOS devices and Si-based lateral power devices. (b) Benchmark of BFOM vs V_{BD} of the HV IGO TFTs in this work, previously reported AOS devices and Si-based lateral power devices.

TABLE I
RECENTLY REPORTED BENCHMARK TABLE OF HV TFTS

Ref.	Channel material	T_{ch} (nm)	W/L (μm)	Gate dielectric	I_{on} (μA)	$R_{on,sp}$ (mΩ·cm²)	V_{BD} (V)	BFOM (kW/cm²)
[4]	PECVD a-Si	50	60/40	150 nm SiN_x	0.6	780000	370	0.18
[5]	PVD IGZO	50	200/50	100 nm SiO_2	8.2	6250	457	33.4
[10]	PVD ITO	6	50/10	20 nm Al_2O_3	110	147	448	1365
[23]	PVD IGZO	20	20/10	150 nm SiO_2	9	1036	91.6	8.1
[24]	PVD ITO	4	5/1	10 nm HfO_2	1615	0.321	156	75813
[27]	PVD IGZO	20	500/50	100 nm SiO_2	15	7200	406	22.9
[28]	PVD IGZO	32.4	2564/28	100 nm Al_2O_3	118	59600	406	2.8
[32]	PVD IGZO	50	100/100	400 nm SiO_2	125	535700	464	0.4
This work	ALD IGO	14	40/5	15 nm HfAlO	794	172	638	2366

can be further improved by the use of high-k dielectrics as passivation layers [26].

Figs. 5(a) and (b) present the $R_{on,sp}$ and BFOM versus V_{BD} benchmarks for IGO devices fabricated by ALD, recently reported state-of-the-art HV TFTs by physical vapor deposition (PVD), and Si-based lateral power devices by plasma enhanced chemical vapor deposition (PECVD) [4], [5], [9], [10], [23], [24], [27], [28], [29], [30], [31], [32]. To improve V_{BD}, channel materials with high critical breakdown electric fields [33], high-quality gate dielectrics, and suitable resistance in the offset region are critical factors in increasing the breakdown voltage [10]. ALD IGO effectively minimizes the resistance of the offset region while maintaining a high V_{BD}, resulting in a superior trade-off relationship between $R_{on,sp}$ and V_{BD} for ALD TFTs. Consequently, a high BFOM of 2.4 MW/cm^2 is obtained [10], [24]. The ALD IGO HV TFTs present an outstanding trade-off relationship between the V_{BD} and $R_{on,sp}$, surpassing the theoretical limit of the single-RESURF Si LDMOS.

Table I summarizes the recently reported benchmarks of HV TFTs using PVD and Si-based lateral power devices by PECVD. The devices in this work achieve a good trade-off in terms of current, breakdown voltage, and on-state resistance.

IV. CONCLUSION

In conclusion, for the first time, high-performance ALD IGO HVTFTs were fabricated with different L_{offset} values. A high carrier mobility of 35 cm^2/V·s, an on-state current of 794 μA, and a low SS of 90 mV/dec were achieved. A record high V_{BD} of 638 V and BFOM of 2.4 MW/cm^2 were also demonstrated. Our results demonstrate the remarkable potential of high-performance ALD AOS HVTFTs for future 3D monolithic integration.

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